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NEURAL BELLMAN OPERATORS

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Neural Bellman Operators construct continuation values and feasible actors for a specified recursive economic objective. We develop an action-level use of information already purchased by an original Bellman certificate. A quadratic minorant and a value enclosure imply a sharp derivative interval on a constrained action domain. A robust feasible replacement then certifies original-policy improvement, regret and objective width without a new value query at the candidate action. The result distinguishes support accuracy, marginal accuracy, implemented-future error and computational cost. We retain complete own-future construction and recursive-risk foundations, with qualified links to stochastic control and nonlinear recursive utility. The numerical application uses the original two-date investment economy with continuous innovations and nested average value at risk. Fifty-six previously acquired actors and an online solver use common original objectives, feasible controls, qualification tolerances and improvement rules; matched-budget representations share the acquired labels. A separately frozen experiment compares neural, matched hinge, quadratic, polynomial, spline and grid representations on 32 new states with three process clocks. Independent rational evaluation, simultaneous population inference and complete cost–value menus distinguish safe policy changes from neural-specific adoption. All 824 accepted changes pass independent gain and containment checks; 102 of 116 own-control mean gains have positive simultaneous lower bounds. Matched hinge and neural decisions coincide, while all complete three-clock neural adoption domains are empty. The resulting information bounds and recursive gain account identify what additional action information a competitive neural decision service must supply.

KEYWORDS: Dynamic programming, Neural approximation, Recursive risk, Policy qualification, Computational economics.

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1. INTRODUCTION

A numerical decision rule is useful when its information changes an economically valuable action at an acceptable total cost. A small approximation residual alone does not answer that question. The distinction becomes especially sharp when a fitted continuation is used inside a constrained recursive decision: its level can be accurately estimated while its action contrasts are wrong, and an expensive certificate can establish the safety of an otherwise uncompetitive policy. This paper studies these issues within Neural Bellman Operators (NBO), which construct and learn continuation operators, return feasible actors, and account for loss under the original recursive objective.

Our application is an investment economy with interacting productive states, state-dependent capacity, continuous innovations and an explicitly nested risk preference. The Bellman recursion, information available to the controller, feasible actions and original objective are fixed throughout each comparison. An implementation must pay for its continuation information, numerical risk integration, training, rejected proposals, qualification and eventual decision. Neural and conventional representations receive the same economic targets and the same certification opportunities. This makes computational adoption an empirical and mathematical question rather than a consequence of a model label.

The contribution. We derive an exact connection between a paid lower quadratic support, an original-value upper enclosure, and the action derivative compatible with that information. The support's slope need not be the derivative of the true Bellman objective. When the support gap is small enough relative to both feasible side lengths, the interior uncertainty scales as its square root; the exact finite-domain bound becomes asymmetric near a capacity boundary. The interval is sharp in its explicitly specified information class. It therefore identifies both what an already completed certificate can buy and what its representation label cannot reveal. For the exact support interface, Supplement, Proposition S.22 matches this upper guarantee with a lower bound for every deterministic or randomized method receiving the same information. The robust proposal is minimax over that stated function class; after the implemented-future allowance, retention is optimal when the possible

guaranteed gain does not cover that allowance. This is an information-class result, not a lower bound for algorithms allowed the additional exact investment formulas.

The interval supplies a feasible policy-improvement mechanism. A robust upper quadratic model produces a small, completely recorded candidate menu. At the realized rounded action, the method checks positive original-policy gain, complete-policy regret and objective width. These checks use the existing payload without querying the original risk at each candidate. An unchanged future controller enters through its explicit error allowance; its numerical error is not silently differentiated. A second service purchases finer support information at the same anchor only under a fixed, costed refinement protocol. The mathematical analysis also separates the precision needed for an action sign from the stricter precision needed for an objective enclosure, and gives a priced stopping dominance condition under a compulsory incremental cost.

The same original-policy advantage also has a recursive account. Supplement, Theorem S.24 propagates nonnegative action advantages computed against a common incumbent continuation through a centered lower risk operator. Its verified subsolutions give lower bounds on the complete-policy gain even when several dates change. The construction requires the inequalities at every relevant state, and a lower bound on the complete risk infimum; one favorable risk witness or a finite sample is insufficient. A stationary extension uses an explicitly invariant Banach space and a discount below one. These statements supply an analytical bridge to a longer original Bellman recursion, not an unexecuted numerical claim or an optimal-convergence guarantee.

These results extend the paper's original operator construction and policy account. The own-future construction uses acquired continuations at every date, returns feasible witnesses, and propagates their errors through the economic recursion. The retained continuous-risk theory states the information interface and lower bounds; the retained stochastic-control and nonlinear recursive- utility results make their additional assumptions explicit. The new experiment implements a two-date specialization. It does not claim that an unexecuted long neural recursion, a nonlinear-utility problem, or a different information set has been numerically solved by that specialization.

Why change the action calculation? The preceding independently evaluated experiment reduced its finite-design marginal-fitting objective in all 28 paired fits, yet increased the enclosed sample mean original economic cost in all 28. Its simultaneous population mean-gain intervals all contained zero. Those results remain in the development companion with their own immutable source and sample. They motivate a distinct question: can the original-risk information already paid for at a proposed action deliver a safer and more valuable action directly? The present study freezes this mechanism before drawing its new states. It reuses the actual stored actors, fully charges their acquisition and fitting, and does not replace the earlier adverse evidence.

What the comparison identifies. Quadratic and polynomial readouts, splines, an ordinary hinge basis, a grid representation and a direct original-model solver are evaluated alongside the ReLU actor. The matched hinge and ReLU representations have the same features and feasible policies; agreement between them is evidence about their common information, not a neural-specific gain. The primary economic question is the change in original policy value. The adoption question additionally charges all acquisition, certification, rejected calculations and complete service work. Population uncertainty is reported separately from descriptive clocks and hypothetical reuse volumes. Every declared cell and comparator is retained, including unsuccessful refinements and empty adoption regions.

The completed experiment accepts 824 original-policy changes, all confirmed by the independent gain and containment checks. The fixed simultaneous family has positive own-control mean-gain lower bounds in 102 of 116 cells. All 768 matched ReLU/hinge primary pairs agree in their decisions and certificates, and all 96 complete three-clock neural adoption menus are empty. The numerical contribution is thus a verified change in original actions and values, together with a complete account of its cost and competitive limits. It does not assign the shared information mechanism to neural representation alone.

Relation to established methods. The classical Lipschitz extension supplies the constructive backend (McShane, 1934). Inexact first-order information and its accuracy requirements have an established optimization literature (Devolder et al., 2014), as

do sharp interpolation inequalities (Taylor et al., 2017). We do not claim generic smooth descent, hinge geometry or bounded convex readouts as neural inventions. The present contribution is the constrained conversion of a specific paid Bellman payload into original-policy gain and qualification, with the implemented-future allowance and a reproducible full-cost comparison. Sections 2–3 give the original operator and construction; Section 6 gives the new information and action mechanism; Section 7 evaluates its economics. Complete proofs for the current main results appear in the technical supplement. The development companion preserves the complete historical theorem and evidence record without substituting it for the current experiment.

2. THE ECONOMIC OPERATOR AND ITS POLICY ACCOUNT

Consider dates $t = 0, \dots, T$ and an invariant compact state domain $X \subset \mathbb{R}^d$. At date t, the action belongs to a nonempty compact set $A_t(x)$. The transition is $F_t(x, a, Z_{t+1})$, the current cost is $c_t(x, a)$, and the terminal cost is $g(x)$. Write $B_0 = 1$ and $B_t = \prod_{j<t} \beta_j$, with $\beta_j > 0$. An implemented policy includes its observation rule, action selection and feasible numerical realization, not just its stored parameters.

For a continuation v, define

$$Q_t(v; x, a) = c_t(x, a) + \beta_t \rho_t(v(F_t(x, a, Z))), \qquad \mathcal{T}_t v(x) = \inf_{a \in A_t(x)} Q_t(v; x, a). \tag{1}$$

Throughout the deterministic comparison, the evaluations ρ_t are monotone and cash invariant on a common domain containing all functions and constant translates used below. They are therefore nonexpansive in the supremum norm. Continuous objectives, effective compact feasible correspondences and measurable minimizers, or measurable arbitrarily near-minimizers with their error charged, are assumed. The stipulated conditional recursion gives Bellman comparison against all admissible adapted controls. In the earlier expected-cost experiments ρ_t is conditional expectation, costs are bounded and innovations have their original continuous uniform law. Section 5 evaluates the same investment technology under an explicitly nested AVaR preference and retains the original continuous innovation law. The direct-cost inference in Section CR is specific to expectation; it is not automatically an inference theorem for nonlinear recursive utility or games.

A Neural Bellman Operator is a construction and evaluation procedure for (1). It works backwards with its own acquired continuation, returns a neural critic and a feasible actor, and records an economic error allowance. Its continuation may be fitted or enclosed by an explicitly paid primitive Bellman calculation; an exact optimal continuation is never supplied as a free oracle. The constructive theorem below uses its own fitted f_{t+1} at every date. In the new two-date application, the terminal function g is an original primitive, and the paid last-date operator $T_1 g$ supplies outward target intervals shared with every conventional comparator. The deployed last-date policy retains its separately proved error. Arbitrary trained networks can be evaluated by the first result below. The constructive backend specifies how a network and its actor are obtained without assuming successful nonconvex optimization.

2.1. *Action-dependent errors and complete-policy loss*

Let $H(a)$ be a true current cost and $\widehat{H}(a)$ its continuation approximation. If $\widehat{a}$ is δ-optimal for $\widehat{H}$ on the same feasible action set, then

$$H(\widehat{a}) - \inf_a H(a) \le \delta + \operatorname{osc}_a\{\widehat{H}(a) - H(a)\}. \tag{2}$$

Indeed, compare $\widehat{a}$ with an arbitrarily near-optimal true action and cancel the approximate objectives. The infimum limit gives the result. Additive continuation levels cancel; action-dependent discrepancies do not. Under a change in future policies, the continuation must be transported to the new economic problem before this inequality licenses reuse. The full development edition retains the original quantitative transport, recursive-domain and economic-application results. No transport bound is presumed valid after an unaccounted change of constraints or preferences.

PROPOSITION 1—One-sided policy sandwich: *Suppose, uniformly in state,*

$$f_t - \mathcal{T}_t f_{t+1} \le u_t, \qquad \mathcal{T}_t^{\pi} f_{t+1} - f_t \le v_t, \qquad \ell_T \le f_T - g \le u_T.$$

Then the actual feasible policy satisfies

$$0 \le J_0^\pi - V_0^* \le \sum_{t<T} B_t(u_t + v_t) + B_T(u_T - \ell_T). \tag{3}$$

The bound is invariant to date-specific constant shifts of the fitted continuations, provided the associated residual accounts are shifted consistently.

The proof is in Supplement, Section 4.1.

A sharper valid residual account for the same continuation and the same deployed actor sharpens knowledge of the policy. It does not retrain that policy or change its realized cost. Conversely, subtracting two upper bounds in (3) does not identify the difference in actual costs. These distinctions govern both the historical comparisons and the new experiments.

3. CONSTRUCTING AND COMPILING A FEASIBLE NEURAL POLICY

Assume the primitive objective has, for feasible state–action pairs, state modulus $A_t^Q = C_{x,t} + \beta_t F_{x,t} L_{t+1}$ and action modulus $D_t = C_{a,t} + \beta_t F_{a,t} L_{t+1}$. Let

$$d_H(A_t(x), A_t(z)) \le K_t^A \|x - z\|_1, \qquad L_t = A_t^Q + D_t K_t^A, \qquad L_T = L_g.$$

The Hausdorff condition and the primitive moduli imply that $\mathcal{T}_t f_{t+1}$ is L_t-Lipschitz whenever f_{t+1} is L_{t+1}-Lipschitz. All moduli and numerical integration procedures are supplied by the economic primitives, not fitted from successful outcomes.

At state nodes $x_{t,i}$ with covering radius h_t, form effective feasible k_t-nets. Evaluate their own-future objectives with certified error at most e_t. Minimize the computed labels and retain the corresponding *original* action $a_{t,i}$. A Borel repair $\Gamma_{t,i}(x)$ is feasible at x and satisfies

$$\|\Gamma_{t,i}(x) - a_{t,i}\|_1 \le K_t^A \|x - x_{t,i}\|_1 + \zeta_t. \tag{4}$$

The allowance ζ_t concerns displacement; feasibility itself may not fail. Define

$$f_t(x) = \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}, \quad i_t(x) = \min \arg\min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}. \tag{5}$$

Return $\pi_t(x) = \Gamma_{t,i_t(x)}(x)$. Terminal labels approximate g with error e_T and use the same cone construction with modulus L_g.

THEOREM 2—Own-future construction with feasible witnesses: *The functions in* (5) *have exact affine–ReLU realizations and Lipschitz moduli* L_t *independent of label error. The returned policy is feasible and measurable. It satisfies*

$$-e_t \le f_t - \mathcal{T}_t f_{t+1} \le D_t k_t + e_t + 2L_t h_t, \tag{6}$$
$$\mathcal{T}_t^{\pi} f_{t+1} - f_t \le e_t + D_t \zeta_t,$$

and hence

$$J_0^{\pi} - V_0^{*} \le \sum_{t<T} B_t (D_t k_t + 2e_t + 2L_t h_t + D_t \zeta_t) + B_T (2e_T + 2L_g h_T). \tag{7}$$

With effective covers, repairs and certified queries at the prescribed accuracy, allocating each nonnegative term a share of a positive economic tolerance gives a terminating finite construction. A fixed numerical precision or a capped finite experimental ladder need not contain that allocation.

The proof is in Supplement, Section 4.2.

The Lipschitz-extension principle is classical (McShane, 1934). A native min-plus envelope and its exact ReLU circuit, using the same labels, witnesses and ties, are *one method with two encodings*. They return the same function and policy. Neither Theorem 2 nor a faster representation proves a uniquely neural capability or a training-success probability for Adam or L-BFGS. The mathematical increment is the complete feasible, own-future construction and implementation account; the empirical question is its work and policy value relative to conventional generators.

4. REFERENCE-CENTERED RISK KERNELS AND COMPLETE-POLICY QUALIFICATION

A datewise worst-case error account can charge incompatible states at successive dates. Under nonlinear recursive preferences, replacing that account by an expectation under the physical transition law is not valid. We instead retain the conditional risk evaluation and center its dual representation at the actual stored NBO continuation. The resulting state-dependent certificate has an original-optimum interpretation and a finite occupation representation. It extends, rather than replaces, the signed qualification of Section 2.

4.1. *The centered operator*

Retain the original augmented economic state, feasible actions, nonnegative discounts, and bounded complete-policy recursions. At each (t, x, a) suppose the original conditional evaluation admits the representation

$$\rho_{t,x,a}(v) = \sup_{q \in \mathcal{Q}_{t,x,a}} \left\{ \int v(y) q(dy) - \psi_{t,x,a}(q) \right\}. \tag{8}$$

Every q is a probability measure on the original successor state space. The representation is assumed on a common bounded-function domain containing all continuations below, with the measurability and selection conditions needed for the original Bellman recursions. This is a hypothesis about the original preference primitive, not permission to replace a static risk objective by a nested one. Conditional risk duality and risk-averse dynamic programming are established principles (Ruszczyński, 2010, 2014). For a nonconvex or non-cash-invariant aggregator lacking (8), the earlier order and modulus results remain applicable.

Fix one qualified parameter θ and suppress its subscript. Its entire sequence of stored continuations is $f = (f_t)_{t=0}^{T}$, and its supplied complete feasible policy is π. Define, using the same f_{t+1} in every term,

$$\begin{aligned} r_t(x, a) &= c_t(x, a) + \beta_t \rho_{t,x,a}(f_{t+1}) - f_t(x), \\ k_t^f(x, a, q) &= \psi_{t,x,a}(q) + \rho_{t,x,a}(f_{t+1}) - \int f_{t+1}\, dq, \\ \mathcal{E}_{t,x,a}^f(z) &= \sup_{q \in \mathcal{Q}_{t,x,a}} \left\{ \int z\, dq - k_t^f(x, a, q) \right\}. \end{aligned} \tag{9}$$

The penalty is nonnegative and has infimum zero, whether or not the supremum in (8) is attained. Direct substitution gives

$$\mathcal{E}_{t,x,a}^f(z) = \rho_{t,x,a}(f_{t+1} + z) - \rho_{t,x,a}(f_{t+1}). \tag{10}$$

In particular, $\mathcal{E}^f(0) = 0$, and $\mathcal{E}^f$ is monotone, cash invariant, and nonexpansive in the uniform norm. Discarding k^f is an upper relaxation, not this identity. Freezing one risk-maximizing kernel at f is in general not even an upper relaxation.

THEOREM 3—State-dependent original-optimum certificate: *Suppose verified functions satisfy* $r_t^-(x,a) \le r_t(x,a)$ *for every original feasible action and* $r_t(x,\pi_t(x)) \le r_t^{\pi,+}(x)$. *Let bounded measurable* L_t, U_t *satisfy*

$$\begin{aligned} L_T &\le g - f_T \le U_T, \\ L_t(x) &\le \inf_{a\in A_t(x)} \{r_t^-(x,a) + \beta_t \mathcal{E}^f_{t,x,a}(L_{t+1})\}, \\ U_t(x) &\ge r_t^{\pi,+}(x) + \beta_t \mathcal{E}^f_{t,x,\pi_t(x)}(U_{t+1}). \end{aligned} \tag{11}$$

Then, at every initial state,

$$f_0 + L_0 \le V_0^* \le V_0^\pi \le f_0 + U_0, \qquad 0 \le V_0^\pi - V_0^* \le U_0 - L_0. \tag{12}$$

The same statements hold at every intervening date and state. They are uniform over a qualified parameter class whenever all premises and the reported bound hold for each member with its own common continuation sequence and complete policy. The certificate permits endogenous states and actions changing at every date; it does not require a finite state space or independent deployment contexts.

With exact residuals and terminal discrepancies, equality recursions in (11) *give* $L_t = V_t^* - f_t$ *and* $U_t = V_t^\pi - f_t$. *With scalar bounds from* (1), *their corresponding constant lower and upper sequences are feasible certificates here. Thus the exact state-dependent recursion is no looser than the signed scalar account. A strictly smaller bound can result without changing a fit, a feasible action, or an economic observation.*

The last assertion concerns the best exact recursion or a verified improvement upon the constant certificates. It is not a claim that every approximately computed L, U improves the scalar account. In implementation, the minimum of valid policy supersolutions and the maximum of valid optimum subsolutions retain validity, so an inferior new bound need not displace the existing one.

The lower inequality requires every original action. A discrete action search without a verified original-menu lower bound cannot establish it. For a fixed action, a feasible risk kernel supplies a lower bound on $\mathcal{E}^f(L)$; an upper bound requires control of the entire risk envelope. Numerical integration, feasible repairs, residual errors and penalty errors belong in these inequalities. If penalty enclosures are

used, lower penalty endpoints enter an upper certificate and upper endpoints enter a lower witness. Approximate dual optimizers are accepted only through these independently checked inequalities.

5. CONTINUOUS-LAW QUALIFICATION IN THE INVESTMENT MODEL

The risk theorem becomes a numerical method only when its inequalities can be constructed from economic primitives. We now supply this construction for the original investment technology with a nested tail-risk preference. States and innovations remain continuous, every feasible action is covered, and the implemented terminal controller rather than a hypothetical exact optimizer determines policy cost. This application joins continuation construction, the original-optimum comparison and a representation comparison on one problem. It does not replace the general NBO construction or the other economic applications.

5.1. *Primitives and the object being approximated*

Let $d \geq 2$ be even, $X = [0,1]^d$, and interpret subscripts cyclically. At dates $t = 0, 1$, the feasible investment is $0 \leq a \leq A(x) = 1/8 + \bar{x}/8$, where $\bar{x} = d^{-1} \sum_j x_j$. Write $b_j = 1/2$ for odd j and $b_j = 1/4$ for even j, and $s_j = (-1)^{j+1}$. The original transition is

$$F_j(x,a,z) = \frac{1}{16} + \frac{x_j}{2} + \frac{x_{j+1}}{8} + \frac{x_j(1-x_{j+1})}{16} + b_j a + s_j z, \qquad z \sim \mathrm{Unif}[-1/32, 1/32]. \tag{13}$$

Define

$$S(x) = \frac{2}{d} \sum_j (x_j - 5/8)^2 + \frac{1}{4d} \sum_j (x_j - x_{j+1})^2 + 2(1/2 - 2\bar{x})_+^2, \tag{14}$$

$$g(x) = S(x) + \frac{2}{d} \sum_j (x_j - 5/8)^2, \qquad h(a) = a^2 + 4a^4.$$

We use $\beta = 15/16$ and, separately, $\alpha \in \{1/2, 3/4\}$. Average value at risk is the upper-tail cost functional

$$\rho_\alpha(v) = \sup \left\{ \int v q \, dP : 0 \leq q \leq (1-\alpha)^{-1}, \int q \, dP = 1 \right\}.$$

Here P is the original uniform innovation law. The objective is recursively defined by $V_2 = g$ and

$$V_t(x) = S(x) + \inf_{0 \le a \le A(x)} \{h(a) + \beta \rho_\alpha(V_{t+1}(F(x,a,\cdot)))\}.$$

It is not the static tail risk of total realized cost. A complete policy value replaces each infimum by the action prescribed by that policy, with the same recursive risk primitive.

Assumptions for the continuous construction. For this subsection, the primitives are exactly (13)–(14); α is fixed before construction; the fitted continuation f is a finite convex function on X; and every numerical lower or upper value is an outward enclosure. Feasible tail densities and feasible actions are retained explicitly. Terminal selection is the measurable, dyadically acquired controller described below, with its fallback included in the error bound. These concrete conditions imply compact feasible graphs, bounded costs, measurable minimizers, and the conditional-risk dual used in Theorem 3. They do not assert that every recursive preference or every trained network has this convex structure.

PROPOSITION 4—Primitive curvature and state supports: *For either risk level and every even d, the one-date objective $Q_1(x,a) = S(x) + h(a) + \beta \rho_\alpha(g(F(x,a,\cdot)))$ is jointly convex on $X \times [0, 1/4]$, with curvature bounded below by*

$$\frac{4 - 5\beta/4}{d} \|v\|_2^2 + 2w^2 \tag{15}$$

in a direction (v,w). Its value V_1 is convex on X and satisfies

$$|V_1(x) - V_1(y)| \le \frac{15}{d} \|x - y\|_1. \tag{16}$$

Let W be the objective evaluated under any fixed feasible risk density, at an anchor (x_0, a_0), and let (p_x, p_a) be its gradient there. For any $\lambda, \nu \ge 0$, set $e = p_a + \lambda - \nu$. Then the following is a global affine minorant of V_1:

$$\ell(y) = W + \lambda(a_0 - A(x_0)) - \nu a_0 - \frac{e^2}{4}$$

$$+\left(p_x-\frac{\lambda}{8d}\mathbf{1}\right)^{\top}(y-x_0). \tag{17}$$

Neither optimal multipliers nor an exact optimizing action are required.

The curvature is obtained from the original nonlinear technology, not imposed on an interpolated value grid. The negative second-order term from $x_j(1-x_{j+1})$ is bounded using $|\partial_j g|\le 10/d$; the stage curvature dominates it. The allowance $e^2/4$ pays for imperfect stationarity. The proof in the Technical Supplement also establishes the global state modulus in (16), including the state-dependent action constraint.

5.2. *An all-action certificate under the original innovation law*

Partition the innovation interval into n equal bins B_i, with midpoints z_i and probabilities $p_i=1/n$. An innovation bin is a set over which an inequality holds, not a replacement point mass in the economic model. Choose nonnegative $q_i\le p_i/(1-\alpha)$ summing to one. These masses define a feasible density constant on each bin. For a fixed initial state and feasible proposal a_0, let W_0 be a lower bound on the objective under this density, obtained from globally valid affine supports of the continuation at the successor midpoints. Let p enclose the derivative of the same supporting objective at a_0. Then

$$\underline{V}_0(x)=\inf_{0\le u\le A(x)}\{W_0+p(u-a_0)+(u-a_0)^2\} \tag{18}$$

is a lower bound on the original Bellman optimum when the supported continuation is V_1. The scalar quadratic is minimized over the full original interval. Interval uncertainty in p is covered by taking the smaller minimum at its two endpoints, since the minimum is concave in p.

For a convex fitted continuation, midpoint supports give the lower risk witness and the larger endpoint value bounds the continuation throughout each bin. For V_1, Proposition 4 supplies midpoint supports and the allowance $15/(32n)$ covers each half-bin. Upper tail risk is the average of the largest $n(1-\alpha)$ upper bin bounds. This average is the exact risk of a dominating step function, hence an upper bound under the continuous law.

THEOREM 5—Constructed original-optimum qualification: *Suppose the supports and bin bounds above are outwardly verified. Suppose further that the complete terminal controller satisfies* $V_1 \le V_1^{\widehat{\pi}} \le V_1 + \varepsilon_1$ *everywhere on* X. *For every queried initial state, the lower value* (18) *and the binwise upper bound* $\overline{J}_0^{\widehat{\pi}}$ *satisfy*

$$0 \le J_0^{\widehat{\pi}}(x) - V_0(x) \le \overline{J}_0^{\widehat{\pi}}(x) - \underline{V}_0(x). \tag{19}$$

For any stored convex continuation f, *the same lower and upper constructions are admissible bounds for the centered-risk comparison of Theorem 3. Their validity does not depend on the representation of* f, *on a successful fit, or on the proposal being an exact minimizer.*

To see the connection, write a verified continuation bracket as $\ell \le V_1 \le u$ and use errors $L = \ell - f$, $U = u - f$. The centered identity gives

$$\rho_\alpha(f) + \mathcal{E}^f(L) = \rho_\alpha(\ell), \qquad \rho_\alpha(f) + \mathcal{E}^f(U) = \rho_\alpha(u).$$

Joint evaluation avoids subtracting and then adding two separately rounded copies of $\rho_\alpha(f)$. The bound (18) uses one feasible risk density to minorize the required all-action infimum; the policy upper bound controls the full risk envelope. Thus the application constructs sufficient inequalities for the original theorem. It does not claim that a non-neural continuation would receive a different validity theorem, or that computing the original-value witness is free.

The terminal controller observes each state coordinate rounded down to 40 binary places. It accepts its floating proposal only after an outward all-action gap of at most 2^{-21} at that observed state; otherwise it uses rational bisection. The original-state transfer is less than $30\,2^{-40}$. Consequently $\varepsilon_1 = 2^{-20}$ covers the complete controller, including observation and fallback. Policy evaluation encloses this controller at every possible successor, not only at the midpoint states used by the integrator.

Well-posed original decisions. The comparison uses a specified conditional nested recursion on the original observation filtration. A sufficient primitive model has a compact state space, nonempty compact feasible sets varying continuously with state, continuous costs and invariant transitions, compact innovations with their original fixed laws, and finite monotone cash-invariant risk evaluations. These

assumptions make the Bellman operator preserve continuous functions. For implemented Borel policies, the risk evaluation must also preserve Borel parameter kernels; the supplement proves this property for expectation and AVaR. Proposition S.10 proves, rather than assumes, the resulting finite-horizon value existence and uniqueness, attained Borel minimizers, comparison against all feasible adapted history policies, and the dated stability bounds. These are classical foundations (Ruszczyński, 2010, 2014), not a claim of a new fixed-point principle. Pointwise transition continuity with an unbounded innovation support is insufficient for arbitrary monetary risk; the exact essential-uniform alternative is stated in the proof.

The original investment primitives satisfy these conditions directly: $F(x,a,z) \in [1/32, 27/32]^d$, $1/8 \le A(x) \le 1/4$, and all costs are bounded. Their joint curvature gives V_1 state strong-convexity modulus $(4 - 5\beta/4)/d$ and root action modulus two. Both dates therefore have unique continuous constrained optimal actors. Proposition S.11 also derives the continuous value gradient with its capacity multiplier, the original-risk root derivative, and the actual observation/feasibility allowance. Differentiating a rounded actor is not used to establish any of these claims. The supplement records the full smooth critic–actor graph and distinguishes its gradient from a frozen-target update.

The separate Brownian recursive-utility bridge has its own existence and verification assumptions. Proposition S.12 supplies the stopped-domain version, charging the original stopping boundary and terminal condition. Its Lipschitz driver can depend on the backward value, in which case cash invariance is not automatic. The continuous-time stability factor and smooth critic graph are therefore verified within that original recursive criterion.

5.3. *Original marginal regularity and the acquired neural critic*

The present computation specializes the preceding investment primitives to $d = 2$. Write $q_x(a) = S(x) + h(a) + \beta\rho_\alpha\{V_1(F(x,a,Z))\}$. Theorem S.4 proves that q_x has a continuous inward derivative on its full feasible interval, is 2-strongly convex, and

has derivative Lipschitz constant

$$L_\alpha = 5 + \beta\left\{\frac{5M_\alpha}{16} + \frac{(4M_\alpha/5)^2\alpha}{(181/64)(1-\alpha)}\right\},$$
$$M_\alpha = \frac{1413}{128} + \beta\left\{\frac{127}{16} + \frac{19625\alpha}{20480(1-\alpha)}\right\}.$$

These constants are derived from the original technology and recursive risk, including binding capacity and an adverse tail touching either innovation boundary. They are not estimates of a fitted critic's curvature.

The retained acquired actor uses an anchored continuation of the form

$$\widehat{R}_\theta(x,v) = \widehat{R}_0(x) + \sum_j \theta_j\{\phi_j(x,v) - \phi_j(x,0)\}, \qquad v = a/A(x).$$

Its known stage cost remains analytic. Paid original-risk value enclosures supply either the value loss or outward finite-difference marginal labels. The complete finite-box fitting procedure and exact optimization-gap audits are in Supplement, Section 3. Their off-design feature-cover theorem states the additional information needed to infer global action accuracy from such training; the earlier experiment did not execute that global cover. The new service below instead acquires and reuses original information at the action actually proposed by the actor.

The resulting complete-policy cost remains

$$J_x(a) = q_x(a) + \eta_x(a), \qquad 0 \le \eta_x(a) \le \delta = \beta 2^{-20}.$$

The allowance is uniform and concerns the original feasible terminal controller. No derivative or convexity of η_x is assumed. A state-dependent value anchor must still be carried when a critic is exported to an earlier Bellman operator: Theorem S.18 gives the full level-and-contrast recursion and its distinct noncash-invariant version. Local action invariance never licenses deleting that recursive level.

6. TURNING A PAID BELLMAN CERTIFICATE INTO A SAFE DECISION

A learned actor can be useful even when its fitted marginal is not a reliable local direction. The original qualification calculation already purchases a lower

supporting objective and an upper original-value enclosure. The missing step is to determine exactly which original action derivatives that information permits, before paying for a second action or a global feature cover. This section gives that conversion and a local policy safeguard. The calculation is available without alteration to every conventional representation in the comparison menu. Its contribution is an action-level use of already paid Bellman information, not a neural-exclusive property of smooth descent. Approximate supporting information and its accuracy–work tradeoff have a classical inexact-oracle interpretation (Devolder et al., 2014). Sharp smooth-convex information bounds also have an established interpolation literature (Taylor et al., 2017). The calculation here states exactly what the saved original-risk support licenses at a capacity-constrained action; it does not claim a new general theory of projected gradient methods.

Fix an original state x, put $A = A(x)$, and write $q = Q_x$ on $[0, A]$. The retained original-model result gives $\mu = 2$ and $L = L_\alpha$ in Theorem S.4; no fitted-critic curvature is used. Suppose a completed original certificate at $a \in [0, A]$ provides numbers W, U, p_-, p_+ and an actual supporting slope $p \in [p_-, p_+]$ for which

$$q(a) \le U, \qquad q(b) \ge W + p(b-a) + \frac{\mu}{2}(b-a)^2 \quad (0 \le b \le A). \tag{20}$$

The slope need not equal $q'(a)$. In particular, the slope of an approximate terminal Bellman minorant is not differentiated as if its optimization error were zero. Put $e = U - W \ge 0$ and $c = (L - \mu)/2$. All these data refer to the original optimal-future objective. A saved upper bound on implemented-policy cost also bounds $q(a)$ and may be used conservatively. Subtracting the future-policy allowance is permitted only when the source formula establishes that the allowance was explicitly added to an otherwise valid q upper bound.

For $r > 0$ define the computable uncertainty function

$$\Psi(e, c, r) = \inf_{0 < h \le r} \{e/h + ch\} = \begin{cases} 2\sqrt{ec}, & c > 0,\ e \le cr^2, \\ e/r + cr, & e > cr^2 \text{ or } c = 0. \end{cases} \tag{21}$$

The first expression has value zero at $e = 0$. When $r = 0$, set $\Psi(e, c, 0) = +\infty$; absence of a feasible direction does not provide an absent derivative bound. A rational or outward implementation can use any feasible h in place of the infimum.

THEOREM 6—Exact derivative information in a paid quadratic support: *Suppose q is continuously differentiable, μ-strongly convex, and has L-Lipschitz derivative, and (20) holds. Then*

$$p_- - \Psi(e, c, A - a) \le q'(a) \le p_+ + \Psi(e, c, a). \tag{22}$$

Intersecting with any separately proved original-gradient enclosure remains valid. For the investment problem $[-7, 7]$ is such an enclosure. For an exact supplied slope p and without that additional gradient information, both endpoints of (22) are sharp in the stated information class: each finite endpoint is attained by a function with constant curvature L, the same support, and $q(a) = U$. At a missing side, the corresponding derivative is unbounded in this class. Thus a method given only these saved data cannot replace a straddling derivative interval by a certified sign using its representation label.

The capacity asymmetry matters. At $a = 0$ the support supplies a lower bound on the inward derivative, which can certify optimality when positive; it does not supply its upper bound. At $a = A$ the opposite statement holds. An arbitrary approximate supporting slope can have a different sign from the actual derivative at a capacity boundary. The supplement gives an exact rational counterexample and verifies the sharpness statements.

6.1. *A replacement which pays no second-action oracle when resolved*

Let $[g_-, g_+]$ be a valid finite version of (22). For a feasible displacement $d \in [-a, A - a]$, define

$$\begin{aligned} \overline{m}(d) &= \max\{g_- d, g_+ d\} + \frac{L}{2} d^2, \\ \underline{m}(d) &= \min\{g_- d, g_+ d\} + \frac{\mu}{2} d^2. \end{aligned} \tag{23}$$

The robust smooth-descent proposal is the minimizer of $\overline{m}$: it is zero if $g_- \le 0 \le g_+$, $-\min\{a, g_-/L\}$ if $g_- > 0$, and $\min\{A - a, -g_+/L\}$ if $g_+ < 0$. The executed finite

menu uses the six multiples $1, 1/2, 1/4, 1/8, 1/16, 1/32$ of this proposal. Among the candidates passing all three tests below, select the one with largest checked lower gain, with ties resolved by the smaller realized action and then the menu index. Retain all six attempted calculations and their costs. A smaller step can pass the objective-width test when the unconstrained robust step cannot. This rule is optimal only over the recorded menu under that certified-gain criterion; it is not a global policy optimizer. The realized action must be feasible after rounding. Recompute the bounds at that realized action; an ideal unrounded minimizer does not license the rounded policy.

The actual future controller is unchanged and satisfies $J_x(b) = q(b) + \eta_x(b)$ with $0 \le \eta_x(b) \le \delta$ from (10). No regularity of η_x is imposed. For the realized $b = a + d$, put

$$\begin{aligned} U_b &= U + \overline{m}(d) + \delta, \\ L_b &= W + \min\{p_- d, p_+ d\} + \frac{\mu}{2} d^2, \\ B &= \min_{s \in \{p_-, p_+\}} \min_{u \in [-a, A-a]} \{W + su + \tfrac{\mu}{2} u^2\}. \end{aligned} \tag{24}$$

A stronger previously verified original-optimum lower bound can replace B by its maximum with that bound.

THEOREM 7—Original-policy gain and qualification from the same payload: *The constructed quantities satisfy*

$$J_x(b) - J_x(a) \in [\underline{m}(d) - \delta, \overline{m}(d) + \delta], \tag{25}$$

$$L_b \le J_x(b) \le U_b, \qquad 0 \le J_x(b) - V_0(x) \le U_b - B. \tag{26}$$

Consequently a strictly negative upper endpoint in (25) *certifies actual economic improvement. When in addition $U_b - B \le \tau$ and $U_b - L_b \le \omega$, the replacement satisfies the same complete-policy regret and objective-width requirements as the incumbent. These three tests use no new original-risk value at b. They do require the original paid payload, its checked smoothness bound, all interval arithmetic, action rounding, and their computational charges. If a test is unresolved, retain the qualified incumbent or enter the separately budgeted original-model fallback.*

A finite sequence of accepted replacements preserves feasibility and qualification, and the sum of certified gains $-\overline{m}(d) - \delta$ lower-bounds the final actual gain. The statement is uniform in state where the premises hold; it is neither a fitted-loss statement nor a population adoption conclusion.

For an interior anchor, an exact supplied support slope, and the exact untruncated interval in (22), the robust upper model is also sharp at each fixed displacement: the worst derivative endpoint and its constant-curvature-L witness attain it. This sharpness does not extend automatically to a wider outward or finite-probe enclosure, or after additional investment-specific information is imposed. Failure of the implemented finite menu therefore records an unresolved calculation; it does not prove that every method using the original formulas must refuse. A neural proposal can locate the first certificate evaluation, but a spline, polynomial, matched hinge, or online method receives exactly the same conversion and safeguard.

6.2. *Decision precision, finite refinement and priced stopping*

The precision of a supporting value and that of its action information have different units. At an interior action with $r = \min\{a, A - a\} > 0$ and $e \le cr^2$, the derivative interval width is at most

$$(p_+ - p_-) + 2\sqrt{2(L-\mu)e}. \tag{27}$$

This is the bound for the exact infimum in (21). A finite list of feasible rational h values remains valid but can be wider; its actual checked radii, including any smallest-step floor, must be used. Thus reducing a support-value error by a factor four reduces this exact component of action uncertainty by only a factor two. Conversely, a certified marginal tolerance ϵ requires both the slope enclosure and the support gap to be allocated explicitly; a small critic fitting loss does not supply either quantity.

Suppose a declared refinement ladder at this same action guarantees $e_k \le e_0 4^{-k}$ and slope width at most $w_0 2^{-k}$, at counted incremental work C_k. These are oracle assumptions to be verified for the implemented original-risk calculations. Put $D = w_0 + 2\sqrt{2(L-\mu)e_0}$. Once the interior condition is reached, the exact-infimum

derivative width is at most $D2^{-k}$. A finite-radius implementation does not inherit this rate without a corresponding radius-approximation guarantee. If $g = q'(a)$ and

$$G_L(a,g) = \max_{d\in[-a,A-a]} \{-gd - Ld^2/2\},$$

the computed unrounded descent has guaranteed q improvement at least $G_L(a,g) - AD2^{-k}$. Hence a true smooth-model margin $G_L(a,g) > \delta + \gamma$ resolves its gain test once $AD2^{-k} < \gamma$, with total actual acquisition work $\sum_{j\le k} C_j$. Qualification width is an additional test. Even at exact derivative information, the Taylor interval can retain width $(L-\mu)d^2/2 + \delta$ at the proposed action. A ladder that cannot meet ω from that information must acquire a candidate value or fall back; it must not declare success from the gain test alone.

There is a narrow but exact priced stopping region. Suppose the current qualified policy has certified original regret at most K, one unit of additional computation costs λ units of original objective, and any allowed continuation of the solver must first pay at least $c_* > 0$ new units. Exclude reuse of that information for other decisions from this single-decision comparison. If $\lambda c_* \ge K$, stopping is weakly optimal against every such continuation: even the exact economic optimum cannot save more than K. This is a dominance rule for a stated compulsory cost, not a general optimal-stopping theorem inferred from historical clock timings. Actual cost intervals, possible reuse, and the complete comparative menu remain part of an adoption study.

7. ORIGINAL-MODEL INVESTMENT DECISIONS AND COMPLETE COSTS

7.1. *Economic design and immutable information*

The experiment uses the original two-date, two-state investment technology of Section 5. Innovations retain their continuous uniform law on $[-1/32, 1/32]$, discounting is $15/16$, and nested AVaR is evaluated at $\alpha = 1/2$ and $3/4$. The action capacity is $A(x) = 1/8 + (x_1 + x_2)/16$ and the direct investment cost is $h(a) = a^2 + 4a^4$. These original primitives, constraints and risk preferences are common to every method. Each final policy must satisfy the same original regret tolerance $\tau = 1/256$ and objective-width tolerance $\omega = 1/8192$.

The 56 stored actors cross two risk levels, two acquisition budgets (5 and 9), seven representations, and value versus marginal fitting losses. The representation menu is quadratic, ReLU, matched ordinary hinge, spline, degree-two polynomial, degree-three polynomial and grid. The original-model online solver has no fitted artifact and remains in the complete menu. The actual acquired models and their acquisition, fitting and finite-box audits retain their preceding immutable identity. There is no new fitting or tuning on the 32 evaluation states. The current scientific source and reporting specification were committed before those states were drawn; the states were committed before any policy result. Under the explicit independent-uniform-bit assumption for the entropy source, the sampling law is the finite 40-bit dyadic law, as stated precisely in Supplement, Section 18. The economic innovations inside each Bellman operator remain continuously distributed.

Each actor receives three complete services: its original qualified control, a *recycled* service using the completed support payload, and an *adaptive* service that can purchase the fixed 256- and 512-cell support refinements at that same anchor. Each preserves the same future controller. There are 58 actor–risk cells, 174 complete service cells and 5,568 primary policy decisions. Three separate serial processes evaluate the fixed first 16 states in rotated cell order, giving 8,352 additional timed decisions. They are descriptive repeat executions of existing states, not additional population observations. The six candidate fractions, realized-action rounding, refinement ladder and tie rule were fixed in advance. No candidate action receives a new original-risk value query from the improvement mechanism. Proposals use the inner 30-bit capacity obtained by rounding the outward lower capacity bound downward; qualification still uses the full original capacity. Derivative recovery uses each feasible side length together with 2^{-j}, $j = 2, \dots, 40$, and intersects with $[-7, 7]$. These finite radii produce valid enclosures; they are not asserted to attain the exact-infimum sharpness or precision rate.

7.2. *Independent economics and population inference*

The independent implementation reads the original primitive model and uses rational terminal optimization and outward continuous-risk integration. It evaluates

every distinct final, incumbent and available raw actor action. A fixed independent 64/128/256/512-cell ladder checks accepted claims; every raw bound and terminal witness is retained. It distinguishes an interval merely compatible with a production certificate from an interval contained within it, and distinguishes a positive gain from confirmation of the entire claimed lower bound. Unresolved checks remain unresolved.

The independently evaluated q is the optimal-future original objective. Its complete-policy interval adds the same separately proved allowance $0 \leq J - q \leq 15/16777216$. This checks the original-policy account for the specified future-controller class; it is not an independent replay of every rounded last-date controller action. Policy regret and action error are reported as justified enclosures or upper bounds, not exact errors inferred from a residual.

The frozen inference family includes all 116 treatment-versus-own-control cells, 232 positive-gain threshold events and 348 signed comparisons against the online control and the matching quadratic value control. The proof in the supplement combines exact binomial inversion with bounded-variable bounds for joint coverage of at least 95%. Numerical enclosures and theorem bounds remain distinguishable from population uncertainty. The inference is conditional on the pre-sample frozen algorithm and inherited information, not conditional on the realized evaluation sample. There is no outcome-selected subset.

7.3. *Complete service work and computational adoption*

Let A denote the complete allocated historical acquisition and fitting cost, H the measured process work outside its nonoverlapping complete cohorts, T one complete cohort of n decisions, and L its measured actor loading cost. The declared scenario for N uses of a stored service is

$$C(N) = A + H + L + \frac{N}{n}(T - L), \qquad N \in \{32, 256, 4096, 65536\}. \tag{28}$$

The calculation includes original information acquisition, tuning and fitting where present, finite-training audits, qualification, candidate arithmetic, rejected candidates, failed paid refinements, actor inference and bookkeeping. Independent

original-model validation, population inference, reporting and publication are separately charged in the research ledger; they are not added twice to a measured decision-service cohort. Shared parent work is allocated in full once to each hypothetical standalone service; those alternative allocations are not added together as realized research time. The primary parent clock excludes its three timing children exactly once. Each timed comparison uses the economic outcomes of the same first 16 states as its clock; the 32-state primary outcomes remain separate.

For a nonnegative price p per unit of measured computation, the complete menu compares $NJ + pC(N)$, with J the sample mean cost per decision. The tables give exact strict-price domains using the independent economic endpoints and recorded clocks, then intersect all traditional competitors and all three timing processes. An additional sensitivity uses the genuinely completed clock from the interrupted earlier attempt and can only reduce that domain. Large N are explicit extrapolation scenarios, not observed throughput or population adoption estimates. Training and certificate reuse therefore enter transparently, without treating an amortization scenario as a new measurement.

The initial execution was interrupted without a terminal parent receipt. Its saved complete cohorts, a complete first timing process and a partial second timing process remain preserved. Before the subsequent complete run, a restart protocol fixed unchanged sources, states and order, required exact decision-field agreement, and forbade new statistical observations. All 9,808 old rows agree on the 16 declared decision fields. The interrupted execution has a measured completed-work lower bound but an unknown total duration; consequently a complete exact research-time total is not asserted. Later schema/reporting corrections preserve their failed attempts, immutable source identities and narrow disclosed changes. None changes a policy outcome.

7.4. *Completed results*

The completed fixed catalogue contains 824 accepted changes among the 3,712 treatment decisions. Every accepted candidate interval is independently contained in its claimed enclosure, and every complete claimed gain lower bound is inde-

TABLE I

SAFE ORIGINAL-POLICY CHANGES IN THE COMPLETE FIXED CATALOGUE

Risk	Service	Changed	Positive means	Refinements	Refusals
1/2	Recycled	212	26/29	0	6
1/2	Adaptive	228	26/29	364	6
3/4	Recycled	182	24/29	0	0
3/4	Adaptive	202	26/29	444	0
Both	All treatments	824	102/116	808	12

Each group contains all 29 actor cells and 928 method–state decisions. "Positive means" counts strictly positive simultaneous population mean-gain lower bounds against the same actor's completed control, within the predeclared joint 95% family. It is not a count of independently sampled actors. Every changed action passes both independent gain and candidate-containment checks. Refinements are completed paid anchor refinements; all 808 attempted refinements completed. Refusals count candidate certificates that fail the stated qualification requirements.

pendently confirmed under the specified future-controller allowance. There are no unresolved accepted checks. These 824 repeated method–state decisions do not supply 824 independent observations: the population analysis has exactly 32 independent initial states. Table I reports every risk–service group, including unchanged actors. The 808 paid anchor refinements and 12 candidate qualification refusals remain in both the records and the cost accounts. The enhancement makes zero additional original-risk queries at candidate actions; its anchor information and all arithmetic still have a measured cost.

The positive population result concerns an action mechanism under its original objective. The lower mean-gain bound is positive in 102 of all 116 treatment cells; the other 14 remain in the catalogue. The observed mean-gain enclosures range from zero to at most 29.006×10^{-6} original cost units across all cells at $\alpha = 1/2$, and from zero to at most 14.855×10^{-6} at $\alpha = 3/4$ (upper endpoints rounded outward). These are small improvements in the cost scale of this economy. They do not establish a representation advantage or the value of an extra unit of computation. The stronger adaptive service accepts 36 more changes than the recycled service across the two risk levels, after purchasing 808 refinements. That aggregate difference describes

TABLE II

THE CONVENTIONAL ONLINE CONTROL ON THE SAME 32 STATES

Risk	J_L	J_U	Regret upper	RMS action upper	Work (s)
1/2	0.655115692	0.655117504	0.000001812	0.001345910	6.8043
3/4	0.664982692	0.664984805	0.000002125	0.001457682	6.4966

J is the original mean cost under the complete policy, enclosed using independent optimal-future evaluation and the stated implemented-future allowance. Regret and root-mean-square action error are upper bounds, not point estimates. Objective and error endpoints are rounded outward. Work is the fully allocated 32-decision primary service cohort, including its share of complete process overhead, rounded for display. These clocks are descriptive measurements; exact rational records and all three 16-state timing processes appear in the complete catalogue.

the two fixed services; it is not a causal decomposition of the value of individual refinements.

The strong original-model solver remains economically material. Table II reports its complete-control benchmarks. For each of the 24 ReLU risk–budget–loss–service cells, the independently enclosed 32-state mean cost lies strictly above the upper endpoint for the same-risk online control. ReLU policy-cost enclosures, including every service and loss, lie within $[0.655349255, 0.655492486]$ at $\alpha = 1/2$ and $[0.665229383, 0.665375326]$ at $\alpha = 3/4$, with the displayed outer endpoints rounded outward. The corresponding fully allocated 32-decision primary cohorts range from approximately 21.10 to 44.13 seconds and 21.86 to 45.28 seconds. These are complete sample comparisons, not a population ordering inferred from 24 independent experiments.

The matched ordinary hinge and ReLU actors agree exactly in all 768 paired primary decisions on the final and incumbent action bit patterns, complete policy endpoints, regret and gain certificates, acceptance, refinement counts and recorded operation counts. The shared hinge geometry therefore cannot be assigned a neural-specific economic contribution. Equally, the safe replacement mechanism is available to all representations, including the online solver, under the same certificate interface.

None of the 348 predeclared signed population comparisons excludes zero in either direction. The own-control population gains thus coexist with unresolved population comparisons across methods. At all four declared reuse volumes, all 96 complete three-clock neural price menus have an empty sufficient domain; their optimistic endpoint domains are empty as well. The 384 separate clock-specific menus and all 28,800 competitor constraints remain available. These finite-sample, measured-clock menus provide no certified neural adoption region. Their emptiness is retained as a result, with the large-volume calculations explicitly treated as reuse scenarios. It does not negate the proved action gains or turn them into a cost advantage. The executed restrictive sensitivity additionally intersects the earlier completed timing process, retaining its original 16-state economics and 7,200 comparator constraints. All 96 four-clock domains remain empty; the earlier process never replaces a less favorable clock.

The development companion contains all 174 policy cells with outward rounded economic endpoints, every one of the 116 own-control contrasts and all 348 timing differences in a fixed catalogue display. Exact rational tables retain the distinctions between raw independent bounds, intersections supported by the policy theorem, and simultaneous population intervals. No row or timing process is selected for a favorable result.

8. ECONOMIC INTERPRETATION AND THE NEXT INFORMATION PROBLEM

The executed mechanism changes original decisions and certifies their value: 824 action changes satisfy the common regret and precision requirements, and all 824 pass the independent gain and containment checks. Positive simultaneous mean-gain lower bounds in 102 of 116 own-control cells establish an economic effect beyond a reduced fitting residual. The effect comes from recovering and using original action information in a paid Bellman certificate. It remains a modest effect in this investment economy.

The complete comparison also identifies what the improvement has not bought. The conventional online control has a strictly lower enclosed sample cost than every ReLU service cell; matched hinge and ReLU decisions are identical. No signed population comparison excludes zero, and none of the 96 complete three-clock

neural menus has a nonempty adoption domain under the declared endpoint comparisons. Neural representation, certified original-policy gain, and competitive complete service cost must therefore remain separate claims. The original NBO program requires an improvement in their combination: continuation and actor information that changes economically valuable choices, a correct recursive policy account, and a competitive acquisition and decision cost.

The information result distinguishes three possible sources of computational value. A representation can locate a promising action; a paid original-model certificate can license its replacement; and reuse can make the complete service attractive at a stated computation price. These are different claims. The first is a learning question, the second is an action-information question, and the third is an economic deployment comparison. A lower fitting residual cannot establish all three. Neither can a successful local replacement establish a neural advantage when an ordinary hinge or a conventional online solver receives the same information and feasible choices.

The future work required by NBO remains substantive. A long learned recursion must retain its continuation level as well as its action contrasts, pay for its own future information, and verify the resulting policy account at every date. The present two-date application supplies an executed instance, not a numerical certificate for those longer recursions. The Brownian nonlinear-utility and FBSDE connections remain conditional on the stated existence, comparison, regularity and information assumptions. Different recursive preferences are not interchangeable economic objectives.

The sharp interval also supplies a concrete research direction. When its uncertainty straddles zero, no representation label resolves the derivative using only the recorded quadratic support and value bound. The matching information bound in Proposition S.22 also rules out a better guaranteed gain from randomizing a proposal at that same exact interface. Its adversary is chosen before the random draw. Additional action information, a different allocation of precision, or exploitable original structure is required. When the derivative sign is resolved but the objective width is not, the remaining obstacle is a different one: curvature uncertainty and the deployed-future allowance impose their own floor. A useful new neural operator

must learn or acquire information that changes these feasible comparisons at lower complete cost, against equally informed strong numerical methods. The current proofs, complete menu and preserved negative results make that question testable without changing the original economic objective.

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